

Problem Set 3

1. (a) $\vec{a} = (\ddot{r} - r\dot{\phi}^2)\hat{n}_n + (2\dot{r}\dot{\phi} + r\ddot{\phi})\hat{n}_t$

$v = ar t$

$a_n = \frac{v^2}{R} = \frac{a^2 t^2}{R}$

(b) $|\vec{a}| = \sqrt{a_n^2 + a_t^2} = \sqrt{\frac{a^2 t^2}{R^2} + a^2} = a \sqrt{\frac{a^2 t^2}{R^2} + 1}$

$\vec{a} = -\frac{a^2 t^2}{R} \hat{n}_n + a t \hat{n}_t$

$\vec{v} = \vec{a} \quad \vec{v} = -\frac{a}{2R^2} t^3 \hat{n}_n + a t \hat{n}_t$

$\vec{r} \perp \hat{n}_n$

$\theta = \arctan \frac{|a_n|}{a_t}$

$\theta = \pi - \arctan \frac{R}{a t^2}$

2. (a) car and ball are both in uniform motion
so for the ball $\uparrow T$ $\downarrow mg$ $\theta = 0$ (FoR as ground)

(b) Take ground as FoR

$\frac{ma}{mg} = \tan \theta$

$\theta = \arctan \frac{a}{g}$

(c)



Take ground as FoR

For the whole $(M+m)g \sin \alpha = (M+m)a$
 $a = g \sin \alpha$

For the ball $ma = mg \sin \alpha$

$\theta = \alpha$

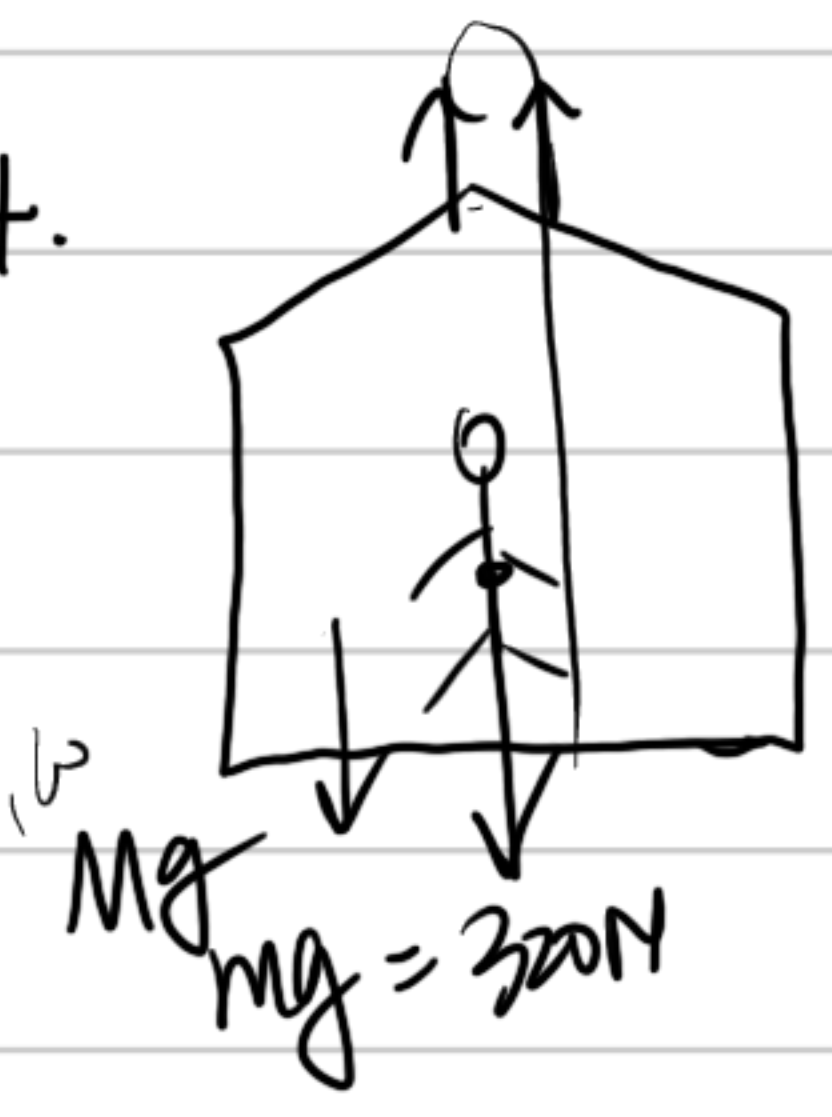
3. $\vec{a} = \frac{\vec{F}}{m} = (-\cos 2t, \frac{3}{2}t^2 - 6t, -3e^{-3t}) \text{ m/s}^2$

since $\vec{v} = \vec{a} \quad \vec{v}(t) = \int \vec{a} \cdot t + \vec{v}(0) = (-\cos 2t, \frac{3}{2}t^2 - 6t, e^{-3t}) + (2, 0, 1)$
 $\frac{dv(t)}{dt} = \vec{a}(t) \quad \int_{v(0)}^{v(t)} dv(t) = \int_0^t \vec{a}(t) dt$

since $\vec{r} = \vec{v} \quad \vec{r}(t) = \int \vec{v} \cdot t + \vec{r}(0) = (2t - \frac{1}{2} \sin 2t, \frac{1}{2}t^3 - 3t^2, -\frac{1}{3}e^{-3t}) + (5, 2, -3)$
 $v(t) = v(0) + (-\cos 2t + 1, \frac{3}{2}t^2 - 6t, e^{-3t} - 1)$

$= (5 + 2t - \frac{1}{2} \sin 2t, 2 + \frac{1}{2}t^3 - 3t^2, -\frac{1}{3}e^{-3t} - \frac{2}{3}) \text{ m}$

4.



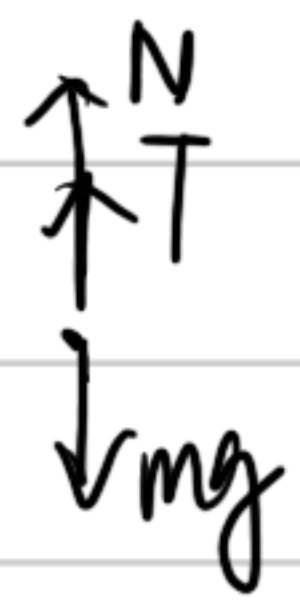
(a) considering the whole

$$2T - (Mg + mg) = (M + m)a$$

$$2 \times 290 \text{ N} - (320 \text{ N} + 160 \text{ N}) = (32 + 16 \text{ kg})a$$

$$a = 0.417 \text{ m/s}^2 \text{ upwards}$$

(b) for the man

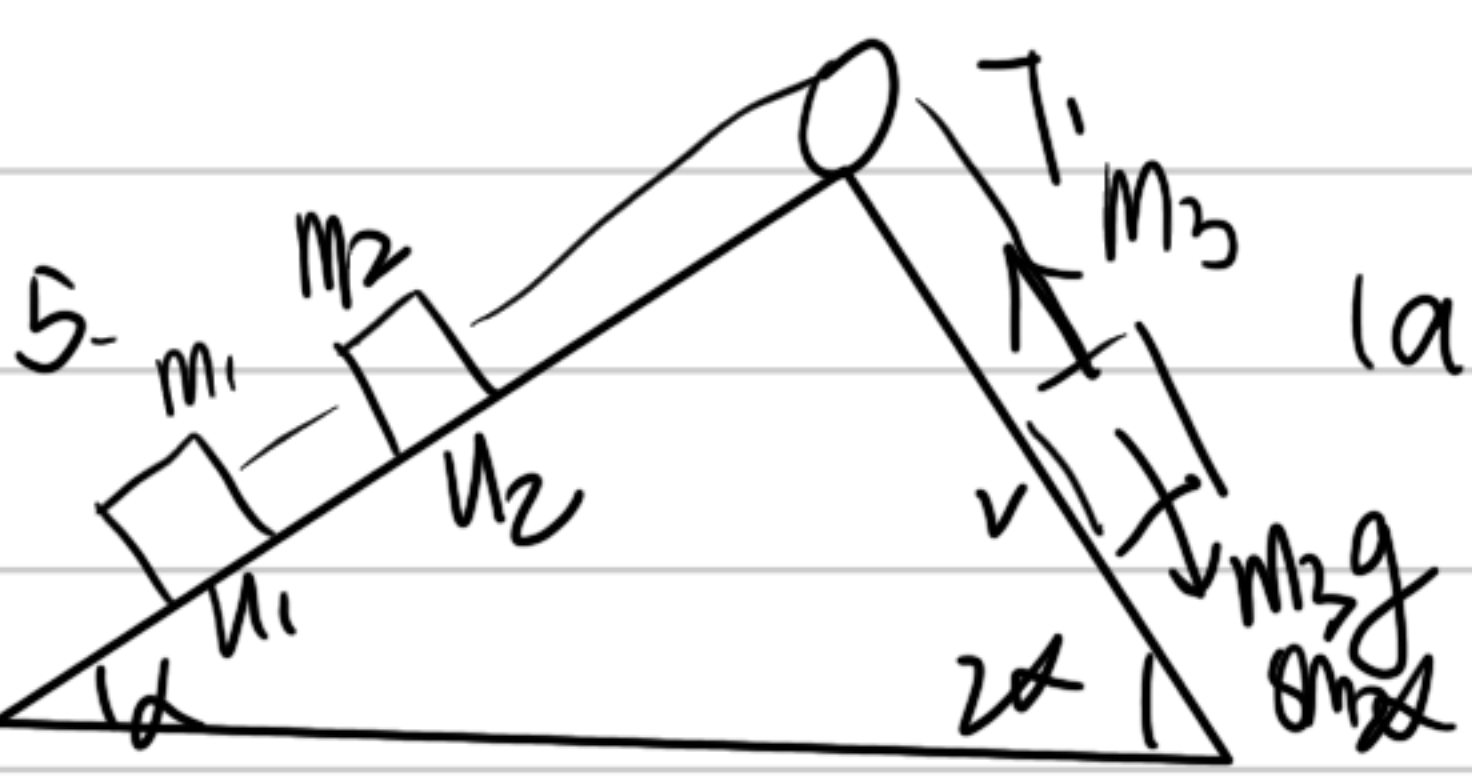


$$N + T - mg = ma$$

$$N = 83.33 \text{ N}$$

According to Newton's 3rd Law.

$$N' = N = 83.33 \text{ N} \quad \text{direction: downwards.}$$



(a) take m_1, m_2, m_3 as a whole

$$m_3 g \sin 2\alpha - (m_1 + m_2) g \sin \alpha - u_1 m_1 g \cos \alpha - u_2 m_2 g \cos \alpha = (m_1 + m_2 + m_3)a$$

$$a = \frac{m_3 g \sin 2\alpha - (m_1 + m_2) g \sin \alpha - u_1 m_1 g \cos \alpha - u_2 m_2 g \cos \alpha}{m_1 + m_2 + m_3}$$

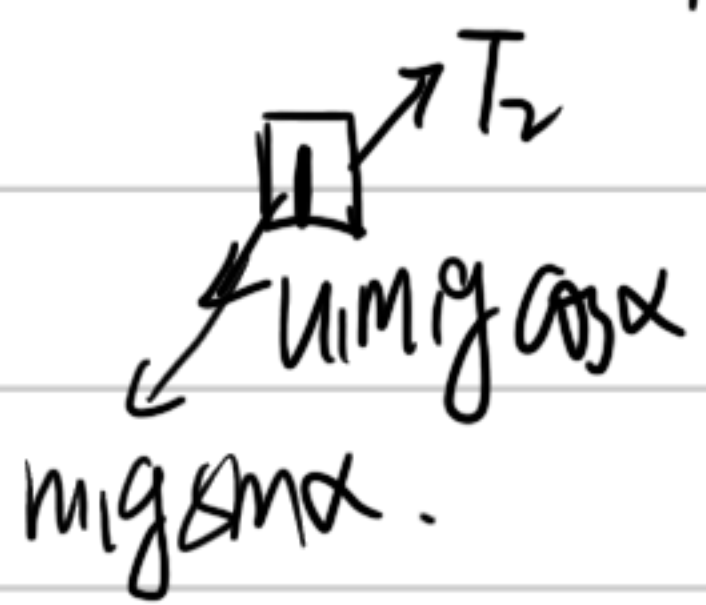
direction: form 2α angle with horizontal line, downward

tension between 2,3, $m_3 g \sin 2\alpha - T_1 = m_3 a$

$$T_1 = m_3 g \sin 2\alpha - m_3 \frac{m_3 g \sin 2\alpha - (m_1 + m_2) g \sin \alpha - u_1 m_1 g \cos \alpha - u_2 m_2 g \cos \alpha}{m_1 + m_2 + m_3}$$

$$= \frac{m_1 m_3 g \sin \alpha + m_2 m_3 g \sin \alpha + m_1 m_3 g \sin \alpha + m_2 m_3 g \sin \alpha + u_1 m_1 m_3 g \cos \alpha + u_2 m_2 m_3 g \cos \alpha}{m_1 + m_2 + m_3}$$

tension between 1,2: $T_2 - u_1 m_1 g \cos \alpha - m_1 g \sin \alpha = m_1 a$

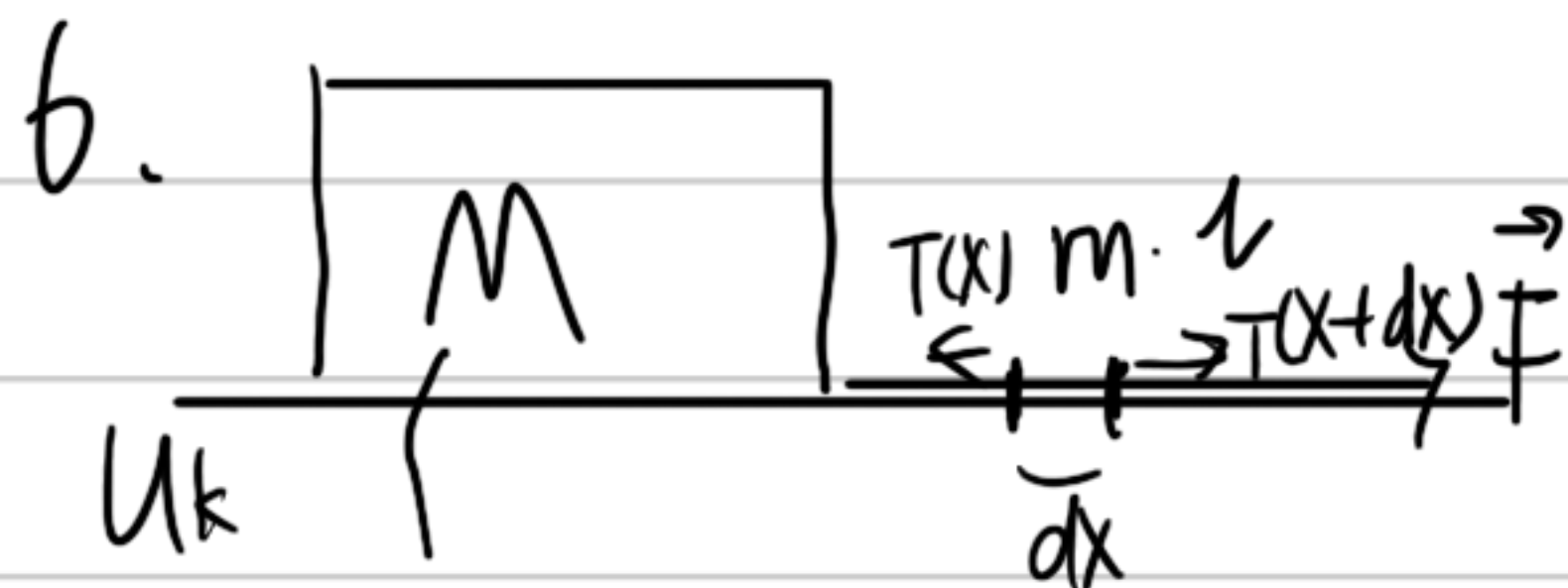


$$T_2 = u_1 m_1 g \cos \alpha + m_1 g \sin \alpha + m_1 a$$

$$T_2 = \mu_1 m_1 g \cos \alpha + m_1 g \sin \alpha + m_1 \frac{m_3 g \sin \alpha - (m_1 + m_2) g \sin \alpha - \mu_1 m_1 g \cos \alpha - \mu_2 m_2 g \cos \alpha}{m_1 + m_2 + m_3}$$

$$= \frac{m_1 m_3 g \sin \alpha + \mu_1 m_1 m_2 g \cos \alpha + \mu_1 m_1 m_3 g \cos \alpha + m_1 m_3 g \sin \alpha - \mu_2 m_1 m_2 g \cos \alpha}{m_1 + m_2 + m_3}$$

$$(b) m_3 g \sin \alpha - (m_1 + m_2) g \sin \alpha - \mu_1 m_1 g \cos \alpha - \mu_2 m_2 g \cos \alpha > 0$$



Take block and rope as a whole

$$F - \mu_k M g = (M + m) a \Rightarrow a = \frac{F - \mu_k M g}{M + m}$$

Choose a point in the rope with distance x from the block
the tension here is $T(x)$

Choose a small segment dx , dx away from $T(x)$ is $T(x+dx)$
for the rope with a length dx $dm = \frac{m}{l} dx$

$$T(x+dx) - T(x) = dm \cdot a$$

$$T(x+dx) - T(x) = \frac{m}{l} dx \frac{F - \mu_k M g}{M + m}$$

$$\frac{T(x+dx) - T(x)}{dx} = \frac{m}{l} \frac{F - \mu_k M g}{M + m}$$

$$\text{when } dx \rightarrow 0 \quad \int_{T(0)}^{T(x)} dT = \int_0^x \frac{m}{l} \frac{F - \mu_k M g}{M + m} dx$$

$$T(x) - T(0) = \frac{m}{l} \frac{F - \mu_k M g}{M + m} x$$

for the block $T(0) - \mu_k M g = M a$

$$T(0) = \mu_k M g + M \frac{F - \mu_k M g}{M + m} = \frac{\mu_k M m g + M F}{M + m}$$

$$T(x) = \frac{\mu_k M m g + M F}{M + m} + \frac{m}{l} \frac{F - \mu_k M g}{M + m} x$$